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15 September 1986
SUPERSEDING
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25 March 1980

MILITARY SPECIFICATION

DRILLING MACHINES, WELL, ROTARY,

SKID- OR TRAILER-MOUNTED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers power-operated drilling machines for drilling vertical wells by imparting pressure to a hollow, rotating, drill bit and drill-bit attachments through which a fluid is pumped to force the loose material at the drill bit to the surface during drilling operations.

1.2 Classification. The well-drilling machines shall be of the following types, as specified (see 6.2):

Type I - Skid-mounted.
Type II - Trailer-mounted.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

FF-B-171	- Bearings, Ball, Annular (General Purpose).
FF-B-185	- Bearings, Roller, Cylindrical; and Bearings, Roller, Self-Aligning.
FF-B-187	- Bearing, Roller, Tapered.
QQ-S-781	- Strapping, Steel, and Seals.
RR-W-410	- Wire Rope and Strand.
UU-S-48	- Sacks, Shipping, Paper.
UU-T-81	- Tags, Shipping and Stock.
ZZ-H-561	- Hose, Rubber, and Hose Assemblies, Rubber, Smooth Bore, Water Suction and Discharge.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-P-40	- Packaging and Packing of Hand Tools.

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MIL-C-104	- Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	- Preservation, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-V-173	- Varnish, Moisture- and Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-P-775	- Packaging of Hose, Hose Assemblies; Rubber, Plastic, Fabric or Metal (Including Tubing); and Fittings, Nozzles and Strainers.
MIL-L-2105	- Lubricating Oil, Gear, Multipurpose.
MIL-F-3541	- Fittings, Lubrication.
MIL-C-3774	- Crates; Wood; Open 12,000- and 16,000-Pound Capacity.
MIL-G-3859	- Grease Guns, Hand-Operated, Lever, Push and Screw-Type
MIL-E-10062	- Engines: Preparation for Shipment and Storage of.
MIL-W-10430	- Welding Rods and Electrodes; Preparation for Delivery of.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-B-11188	- Batteries, Storage: Lead-Acid.
MIL-R-12144	- Reflector, Indicating, Clearance.
MIL-C-13486/12	- Cable, Special Purpose, Electrical: Low Tension, Heavy Duty Multiple Conductors Unshielded.

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MIL-P-13983 - Paint, Temporary, Lusterless, Gasoline Removable.
MIL-G-18458 - Grease, Wire Rope-Exposed Gear.
MIL-L-21260 - Lubricating Oil, Internal Combustion Engine, Preservative and Break-In.
MIL-T-22085 - Tape, Adhesive, Preservation and Sealing.
MIL-G-23827 - Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-P-46093 - Primer Coating, Synthetic (for Brake Drums).
MIL-A-46153 - Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package.
MIL-C-46168 - Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-B-46176 - Brake Fluid, Silicone, Automotive, All-Weather, Operational and Preservative.
MIL-E-52649 - Engine Cold Starting Aids, Ether Fuel Primers.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-281 - Automobiles, Trucks, Truck-Tractors, Trailers and Trailer Dollies; Preservation and Packaging of.
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-781 - Reliability Design Qualification and Production. Acceptance Tests: Exponential Distribution.
MIL-STD-889 - Dissimiliar Metals.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.
MIL-STD-1410 - Method for Selection of Industrial Engines for End Item Application.
MIL-STD-1472 - Human Engineerig Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474 - Noise Limits for Army Materiel.
MS35000 - Battery, Storage, Lead-Acid, Water-Proof.

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MS35647	- Padlocks, Key Operated.
MS35829	- Hinge, continuous (Piano) .075 Thickness, .187 and .0250 Pin Dia.
MS35844	- Plug, Machine Thread, Magnetic: Drain.
MS39061	- Switch, Toggle - SPST, SPDT, 24 Volt, DC, 25 Amp (Waterproof).
MS49006	- Plugs, Pipe, Magnetic-Headless; Aluminum, Iron or Steel.
MS51330	- Stop Light - Tail Light, Vehicular - 24 Volt, Blackout Tail, Blackout Stop.
MS75020	- Connector, Plug, Electrical - 12 Contact, Intervehicular, 28 Volt, Waterproof.

(Copies of specifications, standards, handbooks, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government documents, and publications. The following other Government documents, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DEPARTMENT OF LABOR (DOL)

Occupational Safety and Health Administration Standards (OSHA)

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

DEPARTMENT OF TRANSPORTATION (DOT)

Federal Motor Vehicle Safety Standards (FMVSS).
Federal Motor Carrier Safety Regulations (FMCSR).

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN PETROLEUM INSTITUTE (API)

Specification 5A - Specification for Casing, Tubing, and Drill Pipe.

(Application for copies should be addressed to the American Petroleum Institute, 1801 K Street, NW, Washington, DC 20037.)

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AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY, INC. (AWS)

D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, 2501 NW, Seventh Street, Miami, FL 33125.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

FORGING INDUSTRY ASSOCIATION (FORGI)

Forging Industry Handbook.

(Application for copies should be adddressed to the Forging Industry Association, 1121 Illuminating Building, 55 Public Square, Cleveland, OH 44113).

FRICTION MATERIALS STANDARDS INSTITUTE (FMSI)

Automotive Data Book.

(Application for copies should be addressed to Friction Materials Standards Institute, Inc., 370 Lexington Avenue, New York, NY 10017.)

INDUSTRIAL FASTENERS INSTITUTE (IFI)

IFI Standards.

(Application for copies should be addressed to the Industrial Fasteners Institute, 1505 East Ohio Building, 1717 East Ninth Street, Cleveland, OH 44114.)

REGULAR COMMON CARRIER CONFERENCE (RCCC)

RCCC Recommended Practices Manual.

(Application for copies should be addressed to the American Trucking Association, Inc., 1616 P Street, NW, Washington, DC 20036.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

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(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15086.)

SOUTHERN PINE INSPECTION BUREAU (SPIB)

Standard Grading Rules for Southern Pine Lumber.

(Application for copies should be addressed to the Southern Pine Inspection Bureau, P.O. Box 846, Pensacola, FL 32594.)

STATE OF CALIFORNIA DEPARTMENT OF MOTOR VEHICLES

California Vehicle Code.

(Application for copies should be addressed to the State of California, Department of Motor Vehicles, 2570 24th Street, Sacramento, CA 95809.)

TIRE AND RIM ASSOCIATES, INC. (TRA)

TRA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

TRUCK-TRAILER MANUFACTURERS ASSOCIATION (TTMA)

TTMA Recommended Practices.

(Application for copies should be addressed to the Truck-Trailer Manufacturers Association, Inc., 2430 Pennsylvania Avenue, NW, Washington, DC 20037.)

WEST COAST LUMBER INSPECTION BUREAU (WCLIB)

No. 16 Standard Grading Rules for West Coast Lumber.

(Application for copies should be addressed to the West Coast Lumber Inspection Bureau, P.O. Box 23145, Tigard, OR 97223.)

WESTERN WOOD PRODUCTS ASSOCIATION (WWPA)

Standard Grading Rules for Western Lumber.

(Application for copies should be addressed to the Western Wood Products Association, 1500 Yeor Building, Portland, OR 97204.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail

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specifications, specifications sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The well-drilling machines shall consist of a drilling unit, a mud-pump unit, and a sectionalized collapsible mast. The drilling and mud-pump units shall be separately mounted on skid-type frames designed for uniform distribution of the load. The machines shall be self-contained and equipped as specified herein.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Any changes or deviations of production drilling machines from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish drilling machines conforming to this specification.

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Threads. All screw threads shall conform to FED-STD-H28 and figures 1, 2, and 3.

3.3.2 Wood. Timbers shall be No. 1 Douglas fir conforming to Standard Grading Rules for West Coast Lumber; No. 1 Western larch conforming to Standard Grading Rules for Western Lumber; or No. 1 Southern pine conforming to Standard Grading Rules for Southern Pine Lumber.

3.3.3 Materiel deterioration prevention and control. The drilling machine shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operation and storage environment to which the drilling machine may be exposed.

3.3.3.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the drilling

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system may be newly fabricated from recovered materials to the maximum extent practicable, provided the drilling system produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the drilling system.

3.4 Environmental conditions.

3.4.1 Operating temperature. The drilling machine shall perform as specified herein at any ambient temperature from +32 to +120 °F without overheating or decline in performance.

3.4.2 Storage temperature. The drilling machine shall be stored at any ambient temperature from +160 to -30 °F without damage or decline in performance when operated within a temperature range above +32 °F.

3.5 Safety. All rotating or reciprocating parts, excluding the kelly bar, and other parts subject to high operational temperatures or electrically energized that may become a hazard to operating personnel shall be insulated or properly guarded. Exhaust and other discharges from the equipment shall be directed so that they do not endanger personnel or obstruct the view of the operator. Fuel tanks shall be located in a manner which will not allow spills or overflows to run onto the engine, exhaust, or electrical equipment. Fuel lines shall be adequately supported and shall be protected against mechanical damage.

Human factors engineering. The drilling machine shall conform to Human Factors Engineering Design Criteria as specified in MIL-STD-1472. Special design emphasis shall be given, but not limited to, 4 (General Requirements), 5.1 (Control/Display Integration), 5.2 (Visual Displays), 5.4 (Controls), 5.5 (Labeling), 5.9 (Design for Maintainability), and 5.13 (Hazards and Safety) of MIL-STD-1472, as applicable.

3.7 Noise limits. The noise produced by the drilling machine shall conform to MIL-STD-1474 requirements when tested in accordance with 4.5.2.6. The provisions of MIL-STD-1474, 4.3 and 4.4 shall be provided if and only if MIL-STD-1474, 5.1.1.2 procedures have been pursued and documented to the satisfaction of the procuring activity and written permission to exceed the 85 dB(A) limit is obtained from the procuring activity. Hazard signs shall conform to MIL-STD-1474, 4.3 and be readable at the distance indicated on the sign.

3.8 Reliability. The upper test mean-time-between-failure (MTBF), O_o, shall be 100 hours and the lower test MTBF, O_1, shall be 50 hours when the drilling machine is tested as specified in 4.5.2.7.

3.9 Maintainability.

3.9.1 Maintenance support. Assemblages or support elements such as technical manuals, repair parts, special tools, or lubrication charts, as specified (see 6.6), shall be adequate to perform the intended function of assisting or conducting maintenance operations when tested as specified in 4.5.2.8.

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3.9.2 Maintenance ratio. When tested in accordance with 4.5.2.8, the drilling machines shall have a maintenance ratio of not more than 0.10. Maintenance ratio is defined as the ratio of the total active maintenance man-hours required (scheduled and unscheduled) to the total operating time. Man-hours for repair of replaced components and scheduled before-and-after operation checks are excluded. A maintenance schedule shall be furnished prior to the start of any testing.

3.10 Weights and dimensions. Weights and dimensions with mast down shall not exceed the following:

Type I, skid-mounted

Drilling unit:

Weight, including mast, less operating equipment	3,500 pounds.
Width	60 inches.
Height, overall	60 inches.
Length, overall	92 inches.

Mud-pump unit:

Weight	2,500 pounds.
Width	60 inches.
Height	45 inches.
Length	78 inches.

Type II, trailer-mounted

Drilling unit:

Weight, including mast, less operating equipment	4,200 pounds.
Width	80 inches.
Height, overall	84 inches.
Length, exclusive of towing bar	92 inches.
Length, overall	166 inches.

Mud-pump unit:

Weight	3,200 pounds.
Width	80 inches.
Height, w/wheel mountings	72 inches.
Length, exclusive of towing bar	78 inches.
Length, overall	166 inches.

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3.11 Performance. The drilling machines shall drill holes 4-7/8 inches in diameter to depths of 600 feet for setting 3-inch-inside-diameter casings; holes 5-7/8 inches in diameter to depths of 350 feet for setting 4-inch-inside-diameter casings; and holes 7-7/8 inches in diameter to depths of 150 feet for setting 6-inch-inside-diameter casings.

3.12 Gasoline engines. For the purpose of selecting two 4-stroke cycle engines of the same make and model, the drilling machine shall be classified as a class II end item in accordance with MIL-STD-1410. The engines, including all systems, components, and accessories, shall conform to MIL-STD-1410, class II, except as otherwise specified herein. The engines shall be furnished with a 12-volt cranking and battery-charging system. The battery-charging system shall be of the alternator type of not less than 35-ampere capacity.

3.12.1 Engine control systems. The engines shall be furnished with control systems in accordance with MIL-STD-1410. The control systems shall include a high-temperature control, low-oil-pressure control, and an over speed control. All controls shall be of the engine-shut-off type. The low oil pressure and the engine overspeed controls shall have a buzzer with a 10-second delay to shutdown. The high temperature control shall have a buzzer with a 60-second delay to shutdown.

3.12.2 Engine operation requirements. The engine operational requirements and maximum engine operating limits shall be in accordance with MIL-STD-1410 under any of the requirements and conditions specified herein.

3.12.3 Governing system. The engines shall be provided with a variable-speed governor, gear-driven from the engine. Provisions shall be made for trimming governed speed setting throughout the entire load range while the engine-driven equipment is in operation. The engine speed at any load between no-load and full rated load shall be not less than rated load speed nor exceed rated load speed by more than 10 percent as specified in MIL-STD-1410.

3.12.4 Cold starting aid. When specified (see 6.2), an engine starting aid conforming to MIL-E-52649 shall be furnished.

3.12.5 Fuel tank. The capacity of the tank shall be sufficient for 5 hours of continuous operation in any operating condition specified herein.

3.12.6 Instrumentation. The following instruments shall be included on an instrument panel:

a. Gage, pressure, lubricating oil.
b. Indicator, cooling liquid temperature.
c. Tachometer.
d. Ammeter, battery-charging.
e. Hourmeter.
f. Indicator, fuel, liquid quantity.

3.12.7 Air cleaners. The engine air cleaners shall be of the heavy duty dry type.

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3.12.8 Oil filters. The engine shall be equipped with full flow type oil filters.

3.13 Transmission. The transmission shall be a 4-speed, selective gear type built into the main housing in such a manner that all speeds are available for hoisting and rotating operations. The gear ratio shall range from 1 to 1 in high gear to not less than 6 to 1 in low gear. The transmission shall also have a reverse gear. All gears shall be oil-bath lubricated. Oil seals shall be provided for the input and output shafts. Transmission oil shall conform to MIL-L-2105.

3.14 Clutch. Power shall be transmitted from the engine to the transmission through a 7-inch (minimum size), overcenter-type, single, drydisk clutch, with the driven disk mounted on the transmission input shaft.

3.15 Drilling unit. The drilling unit shall consist of a steel framework upon which is mounted the drill-head assembly, hoisting assembly, collapsible mast, power unit, and operating controls.

3.15.1 Skid frame. The drill-head assembly, hoisting mechanism, mast, power unit, and a toolbox shall be bolted to and supported by a structural steel frame consisting of channel or I-beam skid members reinforced with crossmembers. Both ends of each skid member shall have the web cut back with flanges welded at the ends so as to produce a rounded surface for ease of skidding. The frame shall be provided with towing rings which shall also serve as lifting attachments as specified in 3.25. Sheave wheels shall be installed and arranged on the drill frames to permit the use of the sand reel and line as a winch for skidding and loading the drilling unit. Anchor rings for tong backup lines shall be provided on both sides of the skid frame.

3.15.2 Drill-head assembly. The drill-head assembly shall include a rotary drive-head, power-feed mechanism, transmission, and clutches.

3.15.2.1 Rotary drive-head. The rotary drive-head shall transfer the rotation from the power source to the kelly bar. All components necessary to conduct continuous drilling using a standard drill rod shall be furnished. The assembly shall be hinged so that it can be unlatched by an adjustable hand-operated lock and permitted to swing aside, giving free space for hoisting or lowering drill rods or setting well casings or screens. The hinges shall be adjustable for both vertical and horizontal wear. A means shall be provided for locking the rotary drive in the full open position.

3.15.2.2 Power-feed mechanism. The power-feed mechanism shall provide down-pressure on the kelly bar to increase the pressure on the drilling bit.

3.15.3 Hoisting assembly. The hoisting assembly shall consist of a main-hoist drum and a sand drum, capable of being operated together or independently, mounted in line with the longitudinal axis of the machine. The main-hoist drum and sand drum shall each have not less than four hoisting speeds, and each drum shall be controlled by individual planetary reduction gears. The brake bands, clutch bands, and gear mechanism of both drums shall be interchangeable. Brake levers

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shall be positive locking. Power shall be transmitted to the main-hoist drum and sand-drum shaft through heat-treated sprockets and roller chain. The main hoisting drum shall have a spooling capacity of not less that 80 feet of 1/2-inch wire rope. The sand drum shall have a capacity of not less than 600 feet of 3/8-inch wire rope. The main-hoist drum and the sand drum shall have a maximum line speed of 325 feet per minute, based on bare drum and an engine speed of 1,800 revolutions per minute (rpm).

3.15.3.1 Main hoist line. The drum shall be wound with 80 feet of 1/2-inch-diameter wire rope conforming to RR-W-410, type I, class 4, construction 4, 8 by 19, improved plow steel, right regular lay, preformed, uncoated. The free end of the rope shall be fitted with a safety clevis.

3.15.3.2 Sand drum line. The drum shall be wound with 600 feet of 3/8-inch-diameter wire rope conforming to RR-W-410, type I, class 2, construction 4, 6 by 19, IWRC, improved plow steel, right regular lay, preformed, uncoated. The free end of the rope shall be fitted with a swivel rope socket having a 2-3/8-inch exploration box connection.

3.15.4 Collapsible mast. The mast, or derrick, shall be collapsible, constructed of steel tubing, and shall consist of four legs reinforced with diagonals and crossmembers, electric-welded together in sections and fitted with connections for bolting the sections together. The mast shall have a gross capacity of 10,000 pounds and shall extend 20 feet above the top of the rotary drive-head when the rotary driving-rod is in the lowered position. Side members of the mast shall be in not less than three sections. The two sides of the mast shall be connected with bolted horizontal and diagonal braces. The mast shall be wired for lights, using armored wire or rubber-and-fabric-insulated wire encased in a loom. The wire shall have detachable fittings for disconnection when the mast is dismantled.

3.15.4.1 Crown block. The mast shall be fitted with a detachable, crown-block assembly having sheaves for spooling of the hoisting and sand lines. Sheaves shall be equipped with self-lubricating bearings. An eye-type anchor shall be attached to the underside of the crown block for anchoring the hoisting line when a traveling block is used to pull casing. A detachable guard shall be provided to prevent the wire lines from jumping off the sheaves.

3.15.4.2 Standpipe. A 1-1/4-inch removable standpipe shall be attached to the right side of the mast and shall extend from a point near the base to about mid-height of the mast. The lower end of the standpipe shall be equipped with a 45 degree elbow and union for connection to the discharge from the mud pump. The upper end shall be equipped with a 180 degree turn for connection to the swivel hose.

3.15.5 Power unit. The power unit shall be as specified in 3.12.

3.15.6 Operating controls. The operating controls shall be grouped so that the operator can control the engine, hoisting drums, and swivel head from the same station and located so he can see the point where the drill stem enters the ground. Operating controls shall be in accordance with 3.6.

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3.16 Kelly bar and swivel. The kelly bar shall be made of heat-treated steel with a minimum diameter of 2-3/8 inches and of a length to accommodate the drill rods. The kelly bar shall be equipped with a drive bushing attached to the rotary drill-head in such a manner that rotary drilling motion may be applied to the drill pipe, leaving the kelly bar free to move vertically. The kelly bar shall be equipped on the lower end with a saver sub and an adapter sub for the drill pipe. The swivel shall be equipped with a bail for connection to the hoisting line safety clevis. The swivel shall have a capacity of not less than 5 tons and shall be equipped with a swivel sub for connecting to the kelly bar. The swivel gooseneck shall have a 1-1/4-inch NPT female connection to accommodate the discharge hose.

3.17 Spider base and slips. A spider base with an adjustable spider of maximum capacity for 6-inch drive pipe, bushings for reducing openings, and slips to hold 2-3/8-inch outside-diameter drill pipe, 2-1/2-inch drive pipe, and 4- and 6-inch standard pipe shall be provided. The base shall be equipped with removable backup posts and anchor rings for tielines to backup tongs.

3.18 Mud-pump unit. The mud-pump unit shall be an independent and self-contained unit consisting of a pump, power unit, coupling system, and discharge and suction lines.

3.18.1 Mud pump. The mud pump shall be the horizontal, duplex, double-acting piston type. The pump shall have a capacity of not less than 60 gallons of mud fluid per minute at a pressure of 300 pounds per square inch (psi) ±25 psi when operating at 75 strokes per minute. The pump shall develop working pressures up to 450 psi. Drainage of liquids shall be made from the pump and line into a container with no change in the mounted position of the pump.

3.18.2 Power unit. The power unit shall be as specified in 3.12.

3.18.3 Coupling system. Power from the power unit shall be transmitted to the pump through clutch and reduction drives. All chain and gear drives shall be oil-bath lubricated. Mechanical adjustments shall be furnished to control the tension on drive chains. Housing shall be dustproof.

3.18.4 Discharge line. The discharge piping shall include a shear-type relief valve, with pins and pinhold settings for 1,000, 850, 600, 500, 400, and 200 psi. A 4-1/2-inch, dial-type pressure gage calibrated for pressures from 0 to 1,000 psi shall be mounted on the pump piping and shall be equipped with a gage saver to prevent mud fluids from entering the gage proper. The discharge piping shall also include a 1-1/4-inch gate valve, or 1-inch globe valve, and an air chamber. Arrangement of valves shall permit the use of stand-pipe and mud-mixing hose either independently or together. Valves shall be provided with double-ended arrows and labeled at each end showing functional result (e.g. open, close), in accordance with 3.6.

3.18.4.1 Discharge hose. The discharge hose shall be 1-1/4-inch inside diameter by 17 feet long, to connect the standpipe to the water swivel. The hose shall be fitted with clamps and standard couplings. Threads shall be NPT.

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3.18.5 Suction line. The suction line shall consist of a steel pipe not less than 2 inches inside diameter, extending from the pump to an accessible location for attachment of the suction hose. The free end of the pipe shall be fitted with a ground-joint-union coupling. A swing-type check valve shall be installed in the line.

3.18.5.1 Suction hose and strainer. The suction hose conforming to ZZ-H-561, type II, shall be not less than 2 inches inside diameter by 20 feet long, in two 10-foot sections, and shall be equipped with couplings and a perforated cylindrical strainer. Threads shall be NPT.

3.18.6 Skid frame. The mud pump, power unit, drives, and toolbox shall be bolted to and supported by a structural-steel frame consisting of channel or I-beam skid members reinforced with crossmembers. Both ends of each skid member shall have the web cut back and the flange welded at the ends so as to produce a rounded surface for ease of skidding. The frame shall be provided with towing rings which shall also serve as lifting attachments as specified in 3.26.

3.19 Bearings. Ball and roller bearings shall be of the antifriction type conforming to FF-B-171, FF-B-185, or FF-B-187. All other bearings used shall be high-grade bronze or babbitt.

3.20 Illumination. The drilling machine shall be provided with a 12-volt lighting system to permit daylight performance to be maintained during night operation. Electric power for the lights shall be obtained from the 12-volt electrical system of the power unit. The system shall include a duplex outlet located on the power unit, necessary switches, wiring, and lighting units. All wiring shall be protected against wearing surfaces and weathering.

3.20.1 Floodlights. The drilling machine shall be furnished with two pairs of floodlights to provide the illumination specified in 3.20. Each pair of lamps shall be controlled by a panel-mounted toggle switch conforming to MS39061, and protected by a fuse.

3.20.2 Storage batteries. Batteries conforming to MIL-B-11188 and to MS35000 shall be furnished. The batteries shall be of a size and number to meet the electrical requirements of the drilling machine. The batteries shall be charged and dry. One filling of electrolyte of the required specific gravity shall be shipped with each battery.

3.20.3 Battery box. A weatherproof steel box not less than 0.0747 inch (14 gage) in nominal thickness shall be provided for the batteries. Vents shall be provided near the top of the box and drains shall be provided in the bottom. The box shall be equipped with built-in restraints constructed so that pressure will be applied to the outer battery case only. When the batteries are installed in the box and with all electrical connections completed and when the battery box lid is in the closed position, there shall be no uninsulated electrical conductor within 1 inch of any point within the battery box at which an electrical potential exists. The box shall be located so that the batteries will be accessible for routine maintenance, removal, and replacement. The battery electrolyte level shall be visible when the battery filler caps are

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removed. The batteries shall be accessible for servicing with battery-servicing aids such as a hydrometer and a syringe. Grommets shall be provided to protect the battery lead cables. All interior surfaces of the box shall be treated with acid-resistant paint. A diagram depicting proper battery cable hook up will be permanently affixed to the inside of the lid.

3.21 Accessories and supplies. The following accessories and supplies shall be furnished with each well-drilling machine:

Accessories

Item	Quantity
BAILER, sectional 3-1/2-inch outside diameter (OD), 20-foot long (two 10-foot sections, threaded), with dart valve and 2-3/8-inch exploration pin.	1 ea
BIT, auger, 10-inch, with 2-3/8-inch exploration pin.	1 ea
BIT, fishtail, 3-7/8-inch, with 2-3/8-inch API regular pin.	1 ea
BIT, fishtail, 5-7/8-inch, with 3-1/2-inch API regular pin.	2 ea
BIT, rock, roller, 3-7/8-inch, long-tooth, for medium formation, with 3-3/8-inch API regular pin.	1 ea
HOSE, mud-mixing, rubber, 300-psi, wire-insert 1-inch inside diameter (ID), 25-foot long, with 1-1/2-inch NPT female fitting one end and 6-inch-long by 1/2-inch-diameter pipe nozzle on other end.	1 ea
INSERT, bit 1/8- by 1/4-inch long (approx.), cast tungsten carbide.	2 lb
OVERSHOT, fishing, die, female, for drill rods and tool joints, with 2-3/8-inch exploration box.	1 ea
OVERSHOT, slip, circulating and releasing, for drill rods and tool joints, with 2-3/8-inch exploration box.	1 ea
PLUG, casing, 2-3/8-inch exploration box to 6-inch NPT pin.	1 ea
PLUG, casing, 2-3/8-inch exploration box to 4-inch NPT pin.	1 ea
PLUG, pipe, 2-3/8-inch exploration box to 4-inch NPT pin.	1 ea
PLUG, pipe, 2-3/8-inch exploration box to 2-inch NPT pin.	1 ea
PLUG, hoisting, swivel, wheel-bumper type, with bail, 2-3/8-inch exploration pin.	1 ea
ROD, drill, 10-foot long, 2-3/8-inch exploration, including both pin and box thread protectors in accordance with figures 1, 2, and 3.	35 ea
ROD, drill, 3-foot long, 2-3/8-inch exploration, including both pin and box thread protectors in accordance with figures 1, 2, and 3.	1 ea
SUB, 2-3/8-inch exploration pin to 2-3/8-inch exploration box.	1 ea
SUB, 2-3/8-inch exploration box to 2-3/8-inch exploration box.	1 ea
SUB, 2-3/8-inch exploration box to 2-3/8-inch API regular box.	1 ea
SUB, 2-3/8-inch API regular pin to 3-1/2-inch API regular box.	1 ea
BIT, rock, roller, 5-7/8-inch, long-tooth, for medium formation, with 3-1/2-inch API regular pin.	6 ea
BIT, rock, roller, 7-7/8-inch, long-tooth, for medium formation, with 4-1/2-inch API regular pin.	1 ea

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Accessories

Item	Quantity
BIT, 3-wing, 3-7/8-inch, with 2-3/8-inch API regular pin.	1 ea
BIT, 3-wing, 5-7/8-inch , with 3-1/2-inch API regular pin.	4 ea
BIT, 3-wing, 7-7/8-inch, with 4-1/2-inch API regular pin.	2 ea
BLOCK, traveling, 10-inch, single-sheave, for 1/2-inch wire rope, with swivel safety clevis.	1 ea
CLAMP, pipe, 2-piece, with four bolts and nuts, for 4-inch ID pipe.	1 ea
CLAMP, pipe, 2-piece, with four bolts and nuts, for 6-inch ID pipe.	1 ea
CLEVIS, safety, with 2-3/8-inch exploration pin.	1 ea
COLLAR, drill, 3-1/8-inch OD, 1-1/4-inch bore, 5-foot long, 2-3/8-inch exploration box to 2-3/8-inch API regular box.	1 ea
COLLAR, drill, 4-1/2-inch OD, 1-1/2-inch bore, 5-foot long, 2-3/8-inch exploration box to 3-1/2-inch API regular box.	1 ea
ELEVATOR, steel, with 7/8-inch-diameter by 18-inch-long links fitted with ring for safety clevis; for 6-inch ID pipe, equipped with removable bushing to adapt elevator to 4-inch and 2-1/2-inch ID pipe.	1 ea
ELEVATOR, center-latch type, for 2-3/8-inch exploration drill rods, complete with 7/8-inch-diameter links and ring for safety clevis, 10-ton capacity.	1 ea
HOOK, safety, 10-ton, swivel, 1-3/4-inch opening, with bail.	1 ea
HOOK, strainer, with 2-3/8-inch exploration pin.	1 ea
HOSE, discharge, 1-1/4-inch ID by 17-foot long, wire inserted, with couplings.	1 ea
SUB, 3-1/2-inch API regular pin to 4-1/2-inch API regular box.	1 ea
SUB, 2-3/8-inch exploration pin to 2-3/8-inch API regular pin.	1 ea
SUB, 2-3/8-inch exploration box to 4-1/2-inch API regular box.	1 ea
SWAB, 4-inch, complete with 2-3/8-inch exploration pin.	1 ea
SWAGE BLOCK, 4-inch ID pipe, with 2-inch NPT female connection.	1 ea
SWAGE BLOCK, for 6-inch ID pipe with 2-inch NPT female connection.	1 ea
TAP, fishing, male, for 2-3/8-inch drill rods, with 2-3/8-inch exploration box.	1 ea
MUD, drilling, 100-lb bag.	10 ea
NAILS, common 8d, steel.	5 pounds
ROD, welding, hard surfacing, 3/4-inch diameter, average hardness of C-54 Rockwell when oxyacetylene welded.	101 pounds
ROPE, manila, 1-inch diameter, 65-foot long.	1 ea
STRAINER, tea, 6-inch diameter, No. 30 mesh screen, with wood handle.	1 ea
TIMBER, sizes as follows:	
2-inch by 10-inch nominal by 6-foot long.	6 ea
4-inch by 4-inch nominal by 5-foot long.	4 ea
WASTE, wiping cloth, colored.	25 pounds

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Supplies

Item	Quantity
BABBITT	5 pounds
BRUSH, glue, 1-3/16-inch diameter.	2 ea
CHAIN, close-link, welded, proof coil, 3/8-inch by 10-foot long, with direct eye slip hook (for 3/8-inch chain) on each end.	1 ea
LUBRICANT, tool joint, zinc base, 10-pound (lb) cans.	5 ea

3.22 Electromagnetic interference. Unless otherwise specified (see 6.2), the electromagnetic interference emission characteristics of the drilling machine shall conform to MIL-STD-461, class III C.

3.23 Mobility.

3.23.1 Skid-mounted (type I). The drilling unit shall be mounted on an auxiliary structural-steel frame to provide not less than 13-1/2-inches of vertical-drilling clearance below the end of the kelly bar when in raised position. The auxiliary frame shall be bolted to the drilling-unit frame and shall also be equipped with skid members. Both ends of each skid member shall have the web cut back and the flange welded at the ends so as to produce a rounded surface for ease of skidding. An auxiliary frame is not required for the mud-pump unit. The drilling machine shall load and unload on a truck, utilizing the sheave arrangement on the drill frame with the sand drum and sand line as a winch.

3.23.2 Trailer-mounted (type II). The drilling unit and the mud pump unit shall be individually mounted on separate, single axle, cart trailers. Each unit shall be mounted on a trailer having a manufacturer's gross vehicle weight rating (gvwr) in compliance with DOT Federal Motor Vehicle Safety Standards, equal to or greater than the actual gross vehicle weight (gvw) of the unit with all components installed. Established component and vehicle ratings shall not be increased to meet the requirements of this specification. When towed by its intended truck, each unit shall comply with all over-the-road legal requirements specified herein, and in addition shall meet the following:

a. Withstand transport over expressways, highways, gravel roads, trails, and off-road-terrain to and from the well drilling sites, including towing by crawler tractor as well as truck prime movers.
b. Negotiate side slopes up to 20 percent.
c. Negotiate creek and stream water fordings up to a minimum of 18 inches deep without prior preparation or after-fording services.

3.23.2.1 Trailer requirements. The minimum requirements of each drilling machine trailer shall be in accordance with the following:

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a. Legal requirements:

(1) DOT Federal Motor Vehicle Safety Standards (FMVSS).
(2) DOT Federal Motor Carrier Safety Regulations (FMCSR).
(3) California Vehicle Code.
(4) DOL Occupational Safety and Health Act (OSHA).

b. Industry standards:

(1)	Society of Automotive Engineers	(SAE)
(2)	Truck-Trailer Manufacturers Association	(TTMA)
(3)	Regular Common Carrier Conference	(RCCC)
(4)	Tire and Rim Association	(TRA)
(5)	Friction Materials Standards Institute	(FMSI)
(6)	Industrial Fasteners Institute	(IFI)
(7)	American Welding Society	(AWS)
(8)	Forging Industry Association	(FORGI)

c. Drilling machine installation. All mounting hardware for each drilling machine furnished by contractor; quantity and locations of fasteners comply with requirements for transportability.

d. Bolts, washers, and nuts. Fasteners of high strength hardened types; SAE J429, grade 8; washers flat, hardened steel, one washer adjacent to the bolt and the nut; prevailing torque type mating nuts per IFI Standards 100 and 101; all bolts positioned for top mounting, except where other methods are approved by the contracting officer; not more than three threads exposed on all securely fastened threaded fasteners.

e. Trailer chassis. Rigidity and strength sufficient to withstand maximum equipment payload, both for dynamic transport and for static equipment operation.

f. Frame assembly. Flat framing having longitudinal frame side rails, inner or perimeter type, reinforced with crossmembers; framing of high strength low alloy steels, minimum 50,000 psi guaranteed yield point (yp), corrosion resisting type. Towing connection of the A-frame type; unless otherwise specified (see 6.2), SAE J847 tow bar eye installed in accordance with SAE J849 height range for military design vehicles; when specified (see 6.2); trailer ball coupler in accordance with SAE J684, class 3, for 2-inch ball diameter for commercial vehicles; twin safety chains per DOT.

g. Trailer support. Minimum 5-each, crank handle operated, trailer screw jacks having fold-away feature; jacks located one on towing A-frame and one on each corner of the trailer frame; front jack travel range 22 to 37 inches measured at tow-eye; corner jacks raise tires off ground and withstand equipment operation; jacks enclosed to prevent entrance of foreign matter when in travel position; positive lock in both vertical operating and horizontal travel positions; maximum crank effort not to exceed 40 pound force at the crank handle; jacks equipped with mud pads, minimum 250 square inches per pad, self-leveling type.

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h. Suspension. Flexible, leaf spring, single axle type for heavy duty on-off road usage; withstand static and dynamic loads imposed; suspension rating - 4,000 pound minimum; compression and rebound bumper stops furnished; antisway components linking road- and curb-side wheel; suspension travel, minimum 6-inch full jounce to rebound. Suspension installation in correct alinement with tow eye and trailer centerline.

i. Axles. A 5,000 pound axle manufacturer highway rating, minimum; 1 each hubometer on road side; spindles of alloy steel, hardened and ground. Tapered roller bearings; cups, bearings, and cones in matched sets from the same domestic bearing manufacturer; grease lubrication or oil lubrication with visual oil level windows; axle hubs per SAE J694; 6 stud, 8-3/4-inch bolt circle type.

j. Tires and wheels.

(1) All components per TRA Yearbook.
(2) Heavy duty demountable, single, disc wheel and tire assemblies.
(3) Rim and wheel disc constructed by the same wheel manufacturer, wheel size 9.75-16.5 wide base, drop center tubeless type having a standard disc offset.
(4) Wheel and tire assemblies interchangeable of same size and load range.
(5) Metal valve caps, c-4 type heat-resistant valve cores.

k. Tires.

(1) Size 12-16.5 wide base, tubeless, singles, load range D (8 pair) minimum, having traction lug treads.
(2) Cold inflation pressure matched to load per TRA, maximum 45 psi.
(3) Tire pressure shall be stenciled in one inch high letters above or near the tire "TP (specify lbs.)." The tire pressure shall be as specified in the TRA yearbook.

l. Balancing. Tires, wheels, hubs, and brake drums inherently in balance to prevent wheel tramp, drumming, and excessive tire wear.

m. Wheel splash.

(1) Wheel splash fender and guards per SAE J682, SAE J683, and RCCC recommended practices "Mud Flap Mounting".
(2) Antisail type splash guards having a minimum of 375 pound/inch tear strength, 3,000 psi ultimate strength; material of fiberglass plastic, polyester, or polyvinyl chloride homopolymers.

n. Rear bumper.

(1) DOT FMCSR.
(2) TTMA recommended practices No. 10.

o. Brakes. SAE Handbook; DOT FMVSS: California Vehicle Code.

p. Service brakes. Unless otherwise specified herein, inertia hitch actuated hydraulic with servo-foundation brakes allowing reverse backing; heavy duty master cylinder, emergency break-a-way, fore and aft travel limit stops, corrosion free linkage mechanism; plumbing per DOT, swivel fitting on axle flex line, protected routing; foundation brake size

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minimum 12x2-inches with heavy-duty heat and water fade resistant linings. Hydraulic brake fluid shall conform to MIL-B-46176. When specified (see 6.2), electric brake system complete with controller for installation on 10,000 gvwr commericial pickup truck; 12-volt system.

q. Chock blocks. Two each chock blocks and 2 each cradles per figures 4 and 5 respectively. One assembly located adjacent to the wheels on each side; chain length adequate for placement fore and aft of each wheel.

r. Vehiclular electrical system.

(1) Figure 6; SAE Handbook, electrical equipment and lighting section SAE J821.
(2) DOT FMVSS and FMCSR.
(3) California Vehicle Code.
(4) TTMA recommended practices No. 9 and No. 14.
(5) RCCC recommended equipment specifications.
(6) Heavy duty waterproof type; protected against physical damage, shock, vibration, environment, and corrosion.

s. System voltage. 12-volt; negative ground per SAE J538.

t. Intervehicular cable. Unless otherwise specified (see 6.2), 12-wire intervehicular cable in accordance with MIL-C-13486/12 with pin-type plug in accordance with MS75020-1. When specified (see 6.2) 5-wire intervehicle connectors per SAE J895, including the truck receptacle end and not less than 36-inches of truck leads with mechanical splicing devices on each lead; instructions for truck installation.

u. Wiring. Per SAE J555; heavy duty, rubber or thermosetting plastic per SAE J558, J878 (minimum size 14), and J1067. Supports at least every 3 feet; cushioned clamps, grommets, and plastic loom to prevent whipping, chaffing, and abrasion.

v. Connectors. Per SAE J560, J561, J858, and J928. Connectors plated or solder dipped. Heat shrinkable terminal sleeves. Application of electrical varnish or silicone lubricant to make waterproof. SAE J560 connector will be used with cable described in SAE J1067.

w. Splices. Mechanical splicing devices, rigidly fastened to mounting members; interior waterproofed with silicone lubricant or compound.

x. Lighting. Heavy duty type bulbs and polycarbonate lenses in all lamps; waterproof and shock mounted.

y. Combination lamps (service).

(1) 4 each furnished, SAE-SIT per SAE J759; 2 turn and clearance outboard, 2 stop and tail inboard.
(2) Flush mount type fitted into TTMA recommended practices No. 14, 4-1/2-inch hole in semitrailer rear member other than bumper; elastomer ring with sealed unit or rubber or plastic housing with sealing lens and double contact, indexing bulb; for 12-volt; 3 wire pigtails, 2 hot plus ground, to insure continuity of circuits.

z. Combination lamps (blackout).

(1) Per MS51330 (military type) 2 each, BO-stop and BO-tail.
(2) Installed on rear inboard of service combination lamps; centered behind 6-inch diameter holes with lenses flush with service lamps; underside shielded to prevent damage from road splash.

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aa. Reflectors. 6 each furnished, 2 front, 2 rear, 2 each side; round type with single or twin metal threaded fasteners in accordance with MIL-R-12144.
bb. Wiring diagram. Plate furnished on trailer near the A-frame similar to SAE J821.

3.23.2.2 Servicing and adjustment. The contractor shall service and adjust each trailer for operational use, including not less than the following:

a. Adjustment of operating components and the engine.
b. Flushing and cleaning of the hydraulic brake system.
c. Adjustment of the brake system.
d. Adjustment of the suspension and axles.
e. Continuity inspection of the vehicular electrical system.
f. Check torque of all wheel lugs and inflation of all tires.
g. Complete lubrication.

3.24 Lubrication.

3.24.1 Lubrication provisions. All surfaces requiring lubrication shall be provided with a means for lubricating. The drilling machine shall operate as specified herein when lubricated with military lubricants (see 6.4)

3.24.2 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541, types I, II, or III. Fittings shall be located in a protected position and shall be accessible to a hand-operated grease gun conforming to MIL-G-3859 with a 10-inch flexible extension. Accessiblility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility. A lubrication chart of permanent construction shall be mounted at the operator station showing location of lubrication fittings and type of lubrication, in accordance with 3.6.

3.24.3 Pressure-release device. A pressure release device shall be provided where the use of pressure lubricating equipment could damage grease seals or other parts.

3.24.4 Filling, draining, and checking provisions. Enclosures such as gearcases and transmission housings that contain a reservoir of lubricants for the lubrication of the parts enclosed shall be equipped with dipsticks, checkplugs not less than 1/2-inch pipe size, or sight gages to determine the level of the lubricant. Each enclosure shall be equipped for filling the enclosure with a lubricant and for draining. The drain outlet shall be fitted with a magnetic drainplug, size conforming to MS35844 or MS49006. The drain outlet shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure when the drilling machine is in its normal position. Integral tubes or troughs may be used to convey the lubricant from the drain outlet to a container.

3.24.5 Lubrication of bearings. All bearings, including sealed bearings, shall be lubricated with grease as follows:

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a. MIL-G-23827 - For ball, roller, and needle bearings, gears, and on sliding and rolling surfaces such equipment as instruments and mechanical control systems; for use on gears, actuator screws and other equipment requiring a lubricant with high load-carrying capacity over a temperature range of -100 to +250 °F.
b. MIL-G-10924 - For lubrication and surface corrosion protection of all automotive equipment operated over the temperature range of -65 to +225 °F. Also used in other applications within this temperature range where grease with oxidation-resistant and corrosion-preventative properties is desirable. not for use on machinery which comes in contact with foods or potable water.

The drilling machine shall be assembled, run-in (broken-in), tested, and delivered with these greases. The bearings shall be cleaned before lubrication, because military greases are not always compatible with other greases. A tag shall be attached in a conspicuous place to indicate which military grease has been used.

3.24.6 Lubrication of exposed gears, drive chains, and wire rope. Prior to performance tests specified in 4.5.2.9.1 and 4.5.2.9.2, all exposed gears, drive chains, and wire ropes shall be lubricated with grease conforming to MIL-G-18458.

3.25 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition I or II, as applicable, with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish.

a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.26 Slinging provisions. Permanently affixed slinging provisions that enable the drilling machine to be lifted in its normal travel position, shall be provided and shall conform to MIL-STD-209. Slinging provisions may also be used as tiedowns when such provisions conform to MIL-STD-209.

3.27 Tiedown provisions. Permanently affixed tiedown provisions that permit fastening the drilling machine to the floor or deck of a transportation medium shall be provided and shall conform to MIL-STD-209. Tiedown provisions may also be used as slinging provisions when such provisions conform to MIL-STD-209.

3.28 Identification marking. The drilling machine shall be identified in accordance with MIL-STD-130. The marking shall be applied to the drilling

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machine on plates conforming to MIL-P-514, type I, style 1, composision C, of type I, grade A, class 1 material. Each plate shall be attached by screws, bolts, or rivets in a location where the plate will be both visible and legible.

3.29 Shipping data plate. A shipping data plate conforming to MIL-P-514, type III, composition C, of type I, grade A, class 1 material shall be provided. In addition, the plate shall show the silhouette of the drilling machine in transport position indicating the center of gravity and the location and capacity of the slinging and tiedown provisions. The plates shall be attached by screws, bolts, or rivets in a location where the plates will be both visible and legible.

3.30 Instruction plates. Each lever, switch, and other type control shall be clearly marked to identify its function and shall include instructions, warnings, and cautions where damage to the machine or danger to personnel could result due to incorrect settings during starting, operation, or shutdown of the drilling machine as specified in 3.6. Instruction plates shall conform to MIL-P-514, type III, composition C of type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a location where the plates will be both visible and legible.

3.31 Treatment and painting. The portions of the drilling machine normally painted internally and externally shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G, as applicable. Unless otherwise specified (see 6.2) top coat color shall be camouflage green 383 conforming to MIL-C-46168.

3.32 Toolbox. A toolbox made of steel not less than 0.0749 inch in nominal thickness and of a size to hold maintenance tools and repair parts (see 6.6), shall be furnished. The toolbox lid shall be secured to one edge of a side by a continuous hinge extending the full length of the box edge. The hinge shall conform to MS35829, dash number 7C. The lid shall be held in the closed position by not less than two spring type fasteners with strikers. A hasp shall be provided to secure the lid to the toolbox and accommodate a lock conforming to MS35647. The toolbox shall be securely mounted in an accessible, protected location permitting the lid to be opened through an arc of 90 degrees. The lid shall remain in the fully open position by the weight of the lid or by holding devices. Provisions shall be made to drain the toolbox in its mounted position.

3.33 Workmanship. The drilling machine shall be free from defects such as incomplete welds, rust, cracks, and other defects that could impair the operation or serviceability of the drilling machine.

3.33.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not damage the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

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3.33.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. (Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading.) Parent materials, weld filler metals and fabrication techniques shall be as required to enable the drilling machine to conform to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 0.1875 inch unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication of the drilling machine shall be at the option of the contractor.

3.33.3 Welders. Before assigning any welder or welding operator to welding work covered by this specification, the contractor shall obtain certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualif-
ication is effective as defined by the particular code:

AWS D1.1, Structural Welding Code, Qualification.
ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

Certification shall be made available for review by the contracting officer or designated representive. Contractors who make only horizontal welds need not qualify welders for "all position welding."

3.33.4 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement.

3.33.5 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member, and shall be in accordance with SAE J492.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve

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the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection comparison (see 4.6).
d. Inspection of packaging (see 4.7).

4.3 First article inspection.

4.3.1 Examination. The first article shall be examined as specified in 4.5.1, in order shown. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article shall be subjected to the tests marked "X" in column 1 of table I, in the order shown. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each drilling machine shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Test. Each drilling machine shall be subjected to the test marked "X" in column 2 of table I. Failure of the test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The drilling machines shall be examined as specified herein for the following defects:

101.	Material not as specified.	3.3
102.	Threads not as specified.	3.3.1
103.	Wood not as specified.	3.3.2
104.	Materials are not resistant to corrosion or deterioration or treated to be made resistant to corrosion or deterioration for the applicable storage and operating environment.	3.3.3
105.	Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.	3.3.3.1
106.	Contractor does not have documentation available for identification of materials, material finishes or treatments.	3.3.3.2

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107.	Used, rebuilt or remanufactured components, pieces or parts incorporated in the drilling system.	3.3.4
108.	Safety devices or guards missing.	3.5
109.	Weights and dimensions not as specified.	3.10
110.	Gasoline engines not as specified.	3.12
111.	Engine control systems not as specified.	3.12.1
112.	Governor not as specified.	3.12.3
113.	Cold starting aid not as specified (when furnished).	3.12.4
114.	Fuel tank not as specified.	3.12.5
115.	Instrumentation not as specified.	3.12.6
116.	Air cleaner not as specified.	3.12.7
117.	Oil filters not as specified.	3.12.8
118.	Transmission not as specified.	3.13
119.	Clutch not as specified.	3.14
120.	Drilling unit not as specified.	3.15
121.	Skid frame not as specified.	3.15.1
122.	Drill-head assembly not as specified.	3.15.2
123.	Rotary drive-head not as specified.	3.15.2.1
124.	Power-feed mechanism does not apply down pressure on the kelly bar.	3.15.2.2
125.	Hoisting assembly not as specified.	3.15.3
126.	Spooling capacity of drums not as specified.	3.15.3
127.	Line speed not as specified.	3.15.3
128.	Main hoist line not as specified.	3.15.3.1
129.	Sand drum line not as specified.	3.15.3.2
130.	Collapsible mast not as specified.	3.15.4
131.	Crown block assembly not detachable or not as specified.	3.15.4.1
132.	Operating controls not as specified.	3.15.6
133.	Kelly bar and swivel not as specified.	3.16
134.	Spider base and slips not as specified.	3.17
135.	Mud-pump unit not as specified.	3.18, 3.18.1
136.	Coupling system not as specified.	3.18.3
137.	Discharge line not as specified.	3.18.4
138.	Discharge hose not as specified.	3.18.4.1
139.	Suction line not as specified.	3.18.5
140.	Suction hose and strainer not as specified.	3.18.5.1
141.	Skid frame not as specified.	3.18.6
142.	Bearings not as specified.	3.19
143.	Illumination not as specified.	3.20
144.	Floodlights not as specified.	3.20.1
145.	Storage batteries not as specified.	3.20.2
146.	Battery box not as specified.	3.20.3
147.	Accessories and supplies missing or not as specified.	3.21
148.	Electromagnetic interference emission characteristic not as specified.	3.22
149.	Skid mounting (type I) not as specified.	3.23.1
150.	Trailer mounting (type II) not as specified.	3.23.2
151.	Trailer requirements (type II) not as specified.	3.23.2.1
152.	Trailer/servicing and adjustments not as specified.	3.23.2.2
153.	Lubrication not as specified.	3.24
154.	Lubrication provisions not as specified.	3.24.1

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155.	Lubrication fittings not as specified.	3.24.2
156.	Pressure-release device not as specified.	3.24.3
157.	Filling, draining, and checking provisions not as specified.	3.24.4
158.	Lubrication of bearings not as specified.	3.24.5
159.	Lubrication of exposed gears, drive chains, and wire rope not as specified.	3.24.6
160.	Fungus and moisture resistance treatment not as specified.	3.25
161.	Slinging provisions not as specified.	3.26
162.	Tiedown provisions not as specified.	3.27
163.	Identification markings not as specified.	3.28
164.	Shipping data plate not as specified.	3.29
165.	Instruction plates not as specified.	3.30
166.	Treatment and painting not as specified.	3.31
167.	Color not as specified.	3.31
168.	Toolbox not as specified.	3.32
169.	Workmanship not as specified.	3.33
170.	Metal fabrication not as specified.	3.33.1
171.	Welding not as specified.	3.33.2
172.	Welders certification not made available upon request.	3.33.3
173.	Bolted connections not as specified.	3.33.4
174.	Riveted connections not as specified.	3.33.5

4.5.2 Tests.

4.5.2.1 Test conditions. Prior to test, the drilling machine shall be lubricated with oils and greases specified in 3.24. Oils shall be those designated for use in the ambient temperature at the place of test.

4.5.2.2 Test schedule.

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TABLE I. Test schedule.

First article	Quality conformance	Test	Test paragraph	Requirement paragraph
1	2	3	4	5
X	-	Engine	4.5.2.9.3	3.12
		Environmental:		
X	-	Operation	4.5.2.3.1	3.4.1
X	-	Storage	4.5.2.3.2	3.4.2
X	-	Safety	4.5.2.4	3.5
X	-	Human factors	4.5.2.5	3.6
X	-	Noise level measurements	4.5.2.6	3.7.1
X	-	Reliability	4.5.2.7	3.8
		Maintainability:		
X	-	Maintenance support	4.5.2.8.1	3.9.1
X	-	Maintenance ratio	4.5.2.8.2	3.9.2
		Performance:		
X	-	Drilling machine	4.5.2.9.1	3.11
X	X	Operation	4.5.2.9.2	3.11
X	-	Electromagnetic interference	4.5.2.10	3.22
		Mobility:		
X	-	Skid-mounted (type I)	4.5.2.11.1	3.23.1
X	-	Trailer-mounted (type II)	4.5.2.11.2	3.23.2
X	-	Slinging provisions	4.5.2.12	3.26
X	-	Tiedown provisions	4.5.2.13	3.27

4.5.2.3 Environmental.

4.5.2.3.1 Operation. During performance testing, measure the temperature of the incoming and discharge cooling media at different ambient temperatures and mathematically extrapolate the operating temperature of the engines for an ambient temperature of 120 °F; an operating temperature for the engine of more than 220 °F shall constitute failure of this test.

4.5.2.3.2 Storage. The drilling machine will be soaked in a suitable test chamber at +160 °F for 12 hours and at -30 °F for 12 hours. After exposure to the high and low storage temperature, remove the drilling machine from the test chamber and simulate drilling a well, operating all levers and equipment to determine damage. Any cracks, breaks or permanent deformation of any component, or failure to perform any function properly, shall constitute failure of this test.

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4.5.2.4 Safety. Safety evaluation shall be accomplished throughout testing. Nonconformance to 3.5 shall constitute failure of this test.

4.5.2.5 Human factors. The drilling machine shall be evaluated throughout testing to determine the human factor characteristics. Nonconformance to 3.6 shall constitute failure of this test.

4.5.2.6 Noise level test. Noise levels shall be measured in accordance with MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, figure 7. As a minimum, noise levels shall be measured when equipment is operating under full load. MIL-STD-1474, 5.1.2.1.4 contours shall be taken at not fewer than 12 equal (horizontal) arc increments; one increment shall include data from the noisiest position. Additionally, the noise level at the typical operating position shall be provided as dB(A) level. Failure to comply with MIL-STD-1474 provisions shall constitute failure of this test.

4.5.2.7 Reliability. Using the MTBF's specified in 3.8, the drilling machine shall be tested as specified in 4.5.2.9 with "accept" and "reject" criteria in accordance with MIL-STD-781, test plan IV C. Failure for the purpose of this test is defined as any malfunction which two operators cannot remedy by adjustment, repair, or replacement within 1 hour (all waiting periods are excluded), utilizing the resources available from the parent unit or organization when deployed and which causes or may cause:

a. Failure to commence operation, cessation of operation or degradation of performance capability below the operating conditions required to continue tests specified in 4.5.2.9.
b. Serious damage to the drilling machine by continued operation.
c. Serious personnel safety hazards.

Damage or wear to the drill bit shall be excluded from consideration in determining failure of the drilling machine. Replacement of the drill bit during drilling operations shall be considered as part of the operating procedures and shall not be construed as maintenance. Nonconformance to 3.8 shall constitute failure of this test.

4.5.2.8 Maintainability.

4.5.2.8.1 Maintenance support evaluation. Maintenance operations required during initial production testing shall be accomplished to determine conformance to 3.9.1. Failure of the support elements to maintain the drilling machine as specified shall constitute failure of this test. Errors or inadequacies in the manuals shall not be considered in assessing the maintenance ratio or the MTBF.

4.5.2.8.2 Maintenance ratio. The maintenance ratio shall be computed during first article testing. Nonconformance to 3.9.2 shall constitute failure of this test.

4.5.2.9 Performance.

4.5.2.9.1 Drilling machine. Set up the drilling machine at the drill site. Drill three holes. Drill the first hole to a depth of 600 feet maintaining a

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hole diameter of 4-7/8 inches or larger. Drill the second hole to a depth of 350 feet maintaining a hole diameter of 5-3/8 inches or larger, and set 4-inch casing to case the entire hole. Drill he third hole to 150 feet maintaining a hole diameter of 7-7/8 inches or larger, and set 6-inch casing to case the entire hole. Pull the casings from the holes. Eight hours of drilling shall be at night using the lights mounted on the drilling machine as the only artificial illumination. Maintain a drilling log and indicate the following:

a. Formations encountered and at what depth.
b. Drilling time.
c. Shutdown and reasons (including service and maintenance).

Submit to the contracting officer a certified reproducible copy of the drilling log. Inability to complete the test, reduced drilling rate at night, hole diameters less than specified, inability to set or pull casing, drilling log incomplete, or any permanent deformation or breakage to components of the drilling machine shall constitute failure of this test.

4.5.2.9.2 Operation. Set up the drilling machine for a hole drilling operation. Start the power plant and perform all the operations required to simulate drilling a hole. All major components and all controls shall be operated to maximum performance. Any malfunction, breakage, or permanent deformation shall constitute failure of this test.

4.5.2.9.3 Engine. Prior to assembling the engine to the drilling machine, the engine shall be tested in accordance with the production engine tests and production control tests of MIL-STD-1410. Nonconformance to the requirements specified herein shall constitute failure of this test.

4.5.2.10 Electromagnetic interference. The drilling machine shall be tested as specified in MIL-STD-461. The contractor shall furnish the contracting officer a report of tests required by MIL-STD-461. Disapproval of the report shall constitute failure of this test.

4.5.2.11 Mobility.

4.5.2.11.1 Skid-mounted (type I). Move the skid-mounted drill unit and mud-pump unit of the drilling machine (type I), a distance of 30 feet over dry sandy soil; then load and unload the units using ramps, on trucks. Motive power shall be supplied by the drill unit utilizing the sheave arrangement on the drill frame with the sand drum and sand line as a winch. Inability to load or unload the units, permanent deformation, or breakage shall constitute failure of this test.

4.5.2.11.2 Trailer-mounted (type II). Tow the type II drilling machine for the distances and under the conditions given below:

a. 50 miles over paved highway, 25 miles at 50 miles per hour (mph), and 25 miles at less than 50 mph.
b. 25 miles over unpaved (dirt) roads, 10 miles at 20 mph and 15 miles at less than 20 mph.
c. 10 miles cross country, 5 miles at 10 mph and 5 miles at less than 10 mph.

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Mobility testing shall be performed with the drilling unit in the travel configuration loaded with all equipment and accessories to be carried on the drilling unit. The mud pump unit shall also be loaded with equipment and accessories to be carried on the mud pump unit. Any breakage, permanent deformation, or malfunction shall constitute failure of this test.

4.5.2.12 Slinging provisions.

4.5.2.12.1 Test procedure.

a. lift the drilling machine and hold it in suspension in its normal travel position using slings that converge not more than 24 feet above the lowest extremity of the drilling machine. Determine the forward angles of application for each provision. Measure the clearance between each sling and the drilling machine. Measure the eye openings and clearance dimensions of the provisions.
b. Restrain the drilling machine by anchoring the main frame and subject each slinging provision to a force of 2.5 times the load it carried when suspended. This force shall be applied in the direction as determined in a. above. An acceptable alternate method of test shall be to lift the drilling machine as in a. above and add weights to the drilling machine until the force on each provision is 2.5 times the initial force. Hold the drilling machine under either of the above loading conditions for not less than 90 seconds.

4.5.2.12.2 Failure criteria. Weld failure, permanent deformation of the drilling machine or any slinging provision, slings converging more than 24 feet above the lowest extremity of the drilling machine, or nonconformance to MIL-STD-209 shall constitute failure of this test.

4.5.2.13 Tiedown provisions.

4.5.2.13.1 Test procedure. With the drilling machine anchored by means other than the tiedown being tested, apply the static loads specified in MIL-STD-209 to each provision. Measure the dimensions of the tiedowns.

4.5.2.13.2 Failure criteria. Weld failure, permanent deformation of the drilling machine or its tiedown provisions, breakage of the provisions, or nonconformance to MIL-STD-209 shall constitute failure of this test.

4.6 Inspection comparison. The Government may select drilling machines at any time during the contract production period and subject these drilling machines to the examination and the tests specified in 4.3 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on drilling machines selected at random from those which have been accepted by the Government and will not include the previously inspected first article drilling machines. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

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4.6.1 Inspection failure. Failure of an inspection comparison drilling machine to meet any requirement specified herein during and as a result of the examination and tests specified in 4.6 shall be cause for rejection of the inspection comparison drilling machine and shall be cause for refusal by the Government to continue acceptance of production drilling machines until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on drilling machines previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all drilling machines accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.7 Inspection of packaging.

4.7.1 First article pack inspection.

4.7.1.1 Examination. Examine the first article pack for the defects specified in 4.7.2.3. Presence of one or more defects shall be cause for rejection.

4.7.1.2 Test. The first article pack of crated machines for level A or B shall be subjected to the rail-impact test in appendix A of MIL-STD-1186. Crates not exceeding 9.5 feet in length shall be subjected to test on each end and each side, a total of 8 impacts. At the conclusion of the test, the contents shall be examined. Any shifting of contents, loosening or breaking of holddowns, ties, stays, blocking or bracing, or any visual damage to the contents shall be cause for rejection.

4.7.2 Quality conformance inspection of pack.

4.7.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.7.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.7.2.3 Examination. Samples selected in accordance with 4.7.2.2 shall be examined for the following defects. AQL shall be 1.0 percent defective.

175. Dissassembly not as specified for level a or level B.
176. Matchmarking information not applied as specified for level A or level B.
177. Unfinished surfaces requiring a contact preservative in accordance with the referenced document not coated for level A or level B.
178. Exposed machined surfaces not coated and wrapped or covered as specified.
179. Engine(s) and engine accessories not preserved in accordance with the
180. Cooling system not preserved and tagged as specified for level A, level B, or commercial.
181. Trailer components not preserved in accordance with referenced document for level A or level B.

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182. Clutches and brakes not preserved as specified for level A.
183. Gears not preserved as specified for level A or level B.
184. Rotary drive rod, slip nut, and chuck not preserved and covered or wrapped as specified for level A or level B.
185. Hydraulic system not preserved as specified for level A or level B.
186. Exposed drive chains not preserved as specified for level A or level B.
187. Wire rope not preserved as specified for level A.
188. Standpipe and piping not preserved and threaded surfaces not protected as specified for level A or level B.
189. Kelly and kelly drive bushing assembly not preserved as specified for level A or level B.
190. Mud pump and piston cups not preserved as specified for level A or level B.
191. Removed hose, hose fittings, and strainers not preserved in accordance with the referenced document for level A or level B.
192. Floodlights, if removed, not preserved as specified for level A or level B.
193. Exposed ends of wiring, outlets, and plug openings not sealed with tape for level A.
194. Accessories and supplies not preserved as specified for level A or level B.
195. Repair parts not preserved in accordance with the referenced document for level A or level B.
196. Maintenance tools not preserved in accordance with the referenced document for level A or level B.
197. Consolidation not as specified for level A or level B.
198. Preservation and packing not in accordance with the referenced document as specified for commercial.
199. Packing not as specified for level A or level B.
101a. Marking missing, illegible, incorrect, or incomplete for level A, level B, or commercial.

5. PACKAGING

5.1 First article pack. The contractor shall furnish a first article pack for examination; and crated drilling machines or crated components shall be furnished for test within the time frame specified (see 6.2) to prove prior to starting production packaging that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination and test shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.5). The first article pack may be accomplished utilizing either the first produced drilling machine or a production drilling machine. If the first produced drilling machine is utilized, any preservation, and packing shall be removed by the contractor at no expense to the Government, when requested by the Government, to facilitate comparison between the first produced drilling machine and the production drilling machine.

5.2 Preservation. Preservation shall be level A, level B, or commercial as specified (see 6.2).

5.2.1 Level A.

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5.2.1.1 Disassembly. Disassembly shall be the minimum necessary to safeguard parts vulnerable to damage, pilferage, and loss, and shall be in accordance with the contractor's recommendations. Removed bolts, nuts, pins, and washers shall be placed in mating parts and secured to prevent their loss. Keys removed shall be secured in the keyway of the pulley or flywheel.

5.2.1.2 Matchmarking. When necessary to facilitate reassembly, small detached components shall have matchmarking information on tags conforming to UU-T-81, type A. The information on the tags shall be with waterproof ink. The tags shall be securely attached to the components so as to prevent damage to the preservation or to the components. Large components or assemblies shall be identically matchmarked with letters or numerals using lusterless white paint conforming to MIL-P-13983.

5.2.1.3 Preservatives. Preservatives specified herein shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116.

5.2.1.4 Unprotected surfaces.

5.2.1.4.1 Unfinished surfaces. Unpainted surfaces of items such as tie rods, adjusting rods, clevises, lubrication fittings, spider base and slips, sprockets, drums, sheaves, bolts, nuts, pins, washers, and screws, or any other unpainted exterior metal surface not specified herein requiring the application of a contact preservative in accordance with MIL-P-116, shall be coated with type P-1 preservative.

5.2.1.4.2 Machined surfaces. Exposed machined surfaces of parts shall be coated with type P-6 or P-11 preservative and wrapped or covered with barrier material conforming to MIL-B-121, type I, grade A, class 2, secured in place with tape conforming to MIL-T-22085, type II.

5.2.1.5 Engine(s). The engine(s) and engine accessories shall be preserved in accordance with level A requirements of MIL-E-10062, as specified for type I or II as applicable, method I, using the water and antifreeze procedure for the coolant system.

5.2.1.6 Trailer. Components of the trailer chassis shall be preserved in accordance with the level A (mobile) requirements of MIL-STD-281.

5.2.1.7 Clutches.

5.2.1.7.1 Disc type. With the cover plates removed, and with the clutch engaged, all interior components of the clutch shall be sprayed with primer conforming to MIL-P-46093. Coating of nonmetallic surfaces shall be coated with type P-1 preservative. Clutch control mechanisms not enclosed shall be coated with type P-1 preservative. After being sprayed, spring-loaded-type clutches shall be blocked in a partially disengaged position to eliminate contact between the disc facings and pressure plates. Snap-over-center and toggle-in-type clutches shall be completely disengaged.

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5.2.1.7.2 Band type clutches and brakes. Clutch and brake drum facings and control mechanisms enclosed in the housings shall be coated with primer conforming to MIL-P-46093. Control mechanism not enclosed shall be coated with type P-1 preservative. The clutch and brakes shall not be operated after application of preservatives.

5.2.1.8 Gears.

5.2.1.8.1 Enclosed gears. Gear housings shall be filled to the operating level with the gear oil conforming to MIL-L-2105 and the gears shall be actuated to assure complete coverage.

5.2.1.8.2 Exposed gears. Exposed gears shall be coated with type P-1 preservative or with primer conforming to MIL-P-46093.

5.2.1.9 Rotary drive rod, slip nut, and chuck. The interior and exterior surfaces of the driving rod shall be coated with lubricating oil conforming to MIL-L-21260, grade 30, and the slip nut and chuck shall be coated with type P-1 preservative. The open ends and the exposed surfaces of the rod shall be covered or wrapped with barrier material conforming to MIL-B-121, type I, grade A, class 2, secured in place with tape conforming to MIL-T-22085, type II.

5.2.1.10 Hydraulic system. The hydraulic fluid supply tank shall be filled with the fluid recommended for operation. The pistons shall be fully retracted into the cylinder and secured. Any exposed portions of the piston rods (ramshafts) shall be coated with type P-6 or P-11 preservative and wrapped or covered with barrier material conforming to MIL-B-121, type I, grade A, class 2, extending the wrap approximately 2 inches on the ram cylinder. The wrap shall be secured in place with tape conforming to MIL-T-22085, type II. The hydraulic control valves shall be secured in the neutral position and preserved as specified for piston rods (ramshafts). Hose shall not be disconnected. Any other exterior metal surface of components of the system requiring the application of a contact preservative shall be coated with type P-1 preservative.

5.2.1.11 Hoisting assembly. The main hoist drum and sand reel drum and the planetary reduction gears shall be coated with type P-1 preservative.

5.2.1.12 Exposed drive chains. Exposed drive chains shall be coated with enough type P-3 or P-9 preservative to assure penetration of the preservative to the inner surfaces of the rollers, pins, and bushings. After the excess preservative has drained, the chains shall be coated with type P-1 preservative. Sprockets shall be coated with type P-1 preservative.

5.2.1.13 Wire rope. The wire rope shall be unwound from the drums and coated with type P-1 preservative. The preservative shall be applied by brushing or by immersion. The drums of the hoists shall be coated with type P-1 preservative and the wire rope shall be rewound on the drums and secured to prevent unwinding.

5.2.1.14 Collapsible mast. Unpainted metal surfaces of the mast shall be coated with type P-1 preservative.

5.2.1.15 Crown block. Unpainted surfaces of the crown block assembly including sheaves, anchor, guard, and traveling block shall be coated with type P-1 preservative.

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5.2.1.16 Standpipe and piping. The interior surfaces of the standpipe, detached piping, and connections shall not be coated with a preservative. Exterior unpainted surfaces, except threaded surfaces, shall be coated with type P-1 preservative. Threaded surfaces shall be coated with lubricating oil conforming to MIL-L-21260, grade 30, and metal thread protectors shall be installed.

5.2.1.17 Kelly. The interior surfaces of the kelly shall be coated with lubricating oil conforming to MIL-L-21260, grade 30. The exterior surfaces shall be coated with type P-1 preservative. External threads shall be protected with metal thread protectors and openings shall be covered with barrier material conforming to MIL-B-121, type I, grade A, class 2. The barrier material shall be secured in place with tape conforming to MIL-T-22085, type II.

5.2.1.17.1 Kelly drive bushing assembly. The kelly drive bushing assembly shall be coated with lubricating oil conforming to MIL-L-21260, grade 30, and wrapped with barrier material conforming to MIL-B-121, type I, grade A, class 2. The barrier material shall be secured in place with tape conforming to MIL-T-22085, type II.

5.2.1.18 Mud pump. The fluid cylinders and valve chambers shall be drained. The interior surfaces of the cylinder liners, the cylinder heads, the valve seats, and valve cover plates shall be coated with lubricating oil conforming to MIL-L-21260, grade 30. The cylinder heads and cover plates shall be replaced and the threaded surfaces of the drainplugs shall be coated with the lubricating oil and replaced. The exposed openings through the piston rod lubricating glands shall be covered with barrier material conforming to MIL-B-121, type I, grade A, class 2, secured in place with tape conforming to MIL-T-22085, type II.

5.2.1.18.1 Piston cups. The piston cups shall be removed from the pistons and the cups shall be wrapped together with barrier material conforming to MIL-B-121, type I, grade A, class 2, secured with tape conforming to MIL-T-22085, type II.

5.2.1.18.2 Hose, hose fittings, and strainers. Hose, hose fittings, and strainers detached from the mud pump shall be preserved in accordance with the level A requirements of MIL-P-775.

5.2.1.19 Lighting system. Floodlights, if removed, shall be preserved individually in a close-fitting box conforming to PPP-B-636, W5c or W6c; cushioning to prevent movement and damage shall be provided. Exposed ends of wiring, outlets, and plug openings shall be sealed with tape conforming to MIL-T-22085, type II.

5.2.1.20 Accessories. Exterior unpainted and threaded surfaces of bailer, bits, block, clamps, collars, clevises, elevators, hook, overshots, plugs, drill pipe, subs, swag block, and tap shall be coated with type P-1 preservative. Metal thread protectors shall be used to protect external threads. The interior surfaces of items such as the bailer and drill bits shall be coated with lubricating oil conforming to MIL-L-21260, grade 30. Openings into the interior of all tools shall be covered with barrier material conforming to MIL-B-121, type I, grade A, class 2, secured in place with tape conforming to MIL-T-22085, type II. The rubber hose shall be coiled to the minimum diameter and secured with cotton tape or twine.

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5.2.1.20.1 Inserts. The inserts shall be placed in a close-fitting box conforming to PPP-B-636, W6c, style optional.

5.2.1.21 Supplies.

5.2.1.21.1 Babbitt, nails, and wiping cloths. Each of the items in the quantities specified (see 3.21) shall be placed in a close-fitting box conforming to PPP-B-636, W6c.

5.2.1.21.2 Strainer. The strainer shall be coated with lubricating oil conforming to MIL-L-21260, grade 30, and wrapped with barrier material conforming to MIL-B-121, type I, grade A, class 2. The barrier material shall be secured with tape conforming to MIL-T-22085, type II. The strainer shall then be placed in a box as specified in 5.2.1.21.1.

5.2.1.21.3 Mud. The drilling mud in the quantities specified (see 3.21) shall be contained in sacks conforming to UU-S-48, type II, style A or B, construction 15-15X.

5.2.1.21.4 Glue brushes. Glue brushes shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3.

5.2.1.21.5 Rope. The rope shall be coiled and the coil shall be secured with cotton tape or twine.

5.2.1.21.6 Welding rods. The welding rods shall be preserved in accordance with level A requirements of MIL-W-10430.

5.2.1.21.7 Chain. Unpainted surfaces of the chain shall be coated with type P-1 preservative.

5.2.1.22 Repair parts. The preservative application criteria and applicable method(s) of preservation of MIL-P-116 shall be used to preserve repair parts.

5.2.1.23 Maintenance tools. Maintenance tools shall be preserved in accordance with the level A preservation and packaging requirements of PPP-P-40.

5.2.1.24 Technical publications. Technical publications shall be preserved together in accordance with MIL-P-116, method IC-1 or IC-3.

5.2.1.25 Miscellaneous components. Miscellaneous components not specifically identified herein requiring protection from corrosion or mechanical damage shall be preserved as specified for components of similar design and construction.

5.2.1.26 Consolidation. Components, accessories, and supplies shall be consolidated in a manner most convenient to accomplish the packing specified herein. Components, accessories, and supplies requiring the protection afforded by a wood box shall be packed in close-fitting boxes conforming to PPP-B-621, class 1, style optional, or PPP-B-601, domestic type, style optional. Strapping shall not be required. The contents shall be blocked, braced, and cushioned within the boxes in accordance with MIL-STD-1186.

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5.2.1.26.1 Repair parts, maintenance tools, and technical publications. When space permits, the repair parts, maintenance tools, and technical publications shall be placed in the toolbox. For mobile packaging, the toolbox lid shall be closed and secured in a manner to prevent pilferage. When space does not permit, the repair parts, maintenance tools, and technical publications shall be consolidated as specified in 5.2.1.26.

5.2.2 Level B. The complete drilling machine, components, accessories, and supplies shall be preserved as specified for level A, except items placed in boxes conforming to PPP-B-636, W6c, shall be placed in boxes conforming to PPP-B-636, class domestic; and with the following additional exceptions.

5.2.2.1 Clutches and brakes. Coating of clutch and brake components with primer shall not be required.

5.2.2.2 Hydraulic system. Coating and wrapping of piston rods shall not be required.

5.2.2.3 Exposed drive chains. Exposed drive chains shall be preserved as specified for level A, except only type P-3 preservative shall be applied.

5.2.2.4 Wire rope. Coating of wire rope shall not be required.

5.2.2.5 Lighting system. Taping of exposed ends of wiring, outlets, and plug openings shall not be required.

5.2.3 Commercial. Each complete drilling machine including the accessories and supplies, except for the engine cooling system, shall be preserved in accordanc with ASTM D 3951. Each engine cooling system shall be filled with a fresh clean solution of 50 percent water and 50 percent antifreeze conforming to MIL-A-46153.A tag shall be attached indicating the type of antifreeze installed and the temperature to which the cooling system can be subjected to before damage.

5.3 Packing. Packing shall be level A, level B, or commercial as specified (see 6.2).

5.3.1 Level A.

5.3.1.1 Type I, skid-mounted. Each complete, type I skid-mounted drilling machine with the preserved components, accessories, supplies, repair parts,maintenance tools, and technical publications shall be packed in one or more crates conforming to MIL-C-104, type I, class 1 or 2, style a or b. The contents shall be arranged within the crate to provide maximum compactness, low center of gravity, good balance, and in a manner that void spaces are held to a minimum. The contents shall be blocked, braced, and anchored within the crate in accordance with MIL-STD-1186 and the appendix to the crate specification. Strapping shall be zinc coated.

5.3.1.2 Type II, trailer-mounted. The type II, trailer-mounted drilling machine shall be prepared for mobile shipment. Components that can be secured to the drilling machine or to the trailer without increasing the cube or interfere with lifting or towing shall be secured in a manner to prevent

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movement, pilferage or damage by the use of the holddowns provided and by the use of strapping conforming to QQ-S-781, class 1, type I or IV, finish B or class 2, finish B. Components that are not secured to the drilling machine or trailer shall be packed together in a crate as specified in 5.3.1.1.

5.3.2 Level B.

5.3.2.1 Type I, skid-mounted. Each complete type I, skid-mounted drilling machine shall be packed as specified in 5.3.1.1, except the crates shall conform to MIL-C-3774, type I, style A and the contents protected with a shroud as specified in the appendix to the crate specification.

5.3.2.2 Type II, trailer-mounted. Each complete, type II, trailer-mounted drilling machine shall be prepared for shipment as specified in 5.3.1.2, except if crates are required, they shall be as specified in 5.3.2.1 and the contents protected with a shroud as specified in 5.3.2.1.

5.3.3 Commercial.

5.3.3.1 Type I, skid-mounted. Each complete type I, skid-mounted drilling machine with the preserved components, accessories, supplies, repair parts, maintenance tools, and technical publications shall be packed in accordance with ASTM D 3951.

5.3.3.2 Type II, trailer-mounted. Each type II, trailer-mounted drilling machine shall be prepared for shipment (mobile) in a manner to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with the carrier rules and regulations. Components, accessories, supplies, repair parts, maintenance tools and technical publications not contained on the trailer-mounted drilling machine shall be packed in accordance with ASTM D 3951.

5.4 Marking.

5.4.1 Military. Marking for military levels of protection (level A and level B) shall be in accordance with MIL-STD-129.

5.4.2 Commercial. Packages, shipping containers, and unboxed drilling machines shall be marked in accordance with ASTM D 3951. In addition, shipping containers shall indicate the weight and cube.

6. NOTES

6.1 Intended use. The drilling machines are intended for use in developing water-producing wells.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type of drilling machine required (see 1.2).
c. Time frame for submission and number of first articles required (see 3.2).
d. When the Government will conduct any or all of the first article examination and tests, the contracting officer should specify which examinations and tests will be conducted by the Government and which examinations and tests shall be conducted by the contractor (see 3.2).

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e. When cold weather starting aid is required (see 3.12.4).
f. Color when other than as specified (see 3.21).
g. When the electromagnetic interference shall be other than as specified (see 3.22).
h. When trailer ball coupler shall be furnished in lieu of tow bar eye (see 3.23.2.1f).
i. When electric brake system is required (see 3.23.2.1p).
j. When 5-wire intervehiclular connectors are required (see 3.23.2.1t).
k. When welding process is not at the contractor's option (see 3.33.2).
l. Time frame required for submission of the first article pack (see 5.1).
m. Degree of preservation and degree of packing required (see 5.2 and 5.3).

6.3. First article. When a first article inspection is required, the item should be a preproduction model. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the document's first article.

6.4 Lubricants. MIL-STD-838, Lubrication of Military Equipment, prescribes the policy for using specification-type products wherever possible and provides specific requirements for potential use of non-standard proprietary products. MIL-STD-838 is implemented by MIL-HDBK-113, Guide for the Selection of Lubricants, Fluids, Preservatives and Speciality Products for Use in Ground Equipment Systems. The contracting officer should note that unless otherwise authorized by the US Army Belvoir Research, Development and Engineering Center (ATTN: STRBE-VF), Fort Belvoir, VA 22060-5606, lubricants, fluids, and greases for ground equipment systems must be restricted to those listed under MIL-HDBK-113, chapter 2. MIL-HDBK-113, chapter 2.

6.5 First article pack. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preserve, pack, and mark the drilling machines in accordance with this specification.

6.6 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of the drilling machines.

6.7 Subject term (Key word) listing.

Drilling machine
Drilling machine, skid-mounted
Drilling macnine, trailer-mounted
Drilling machine, well, rotary

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6.8 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

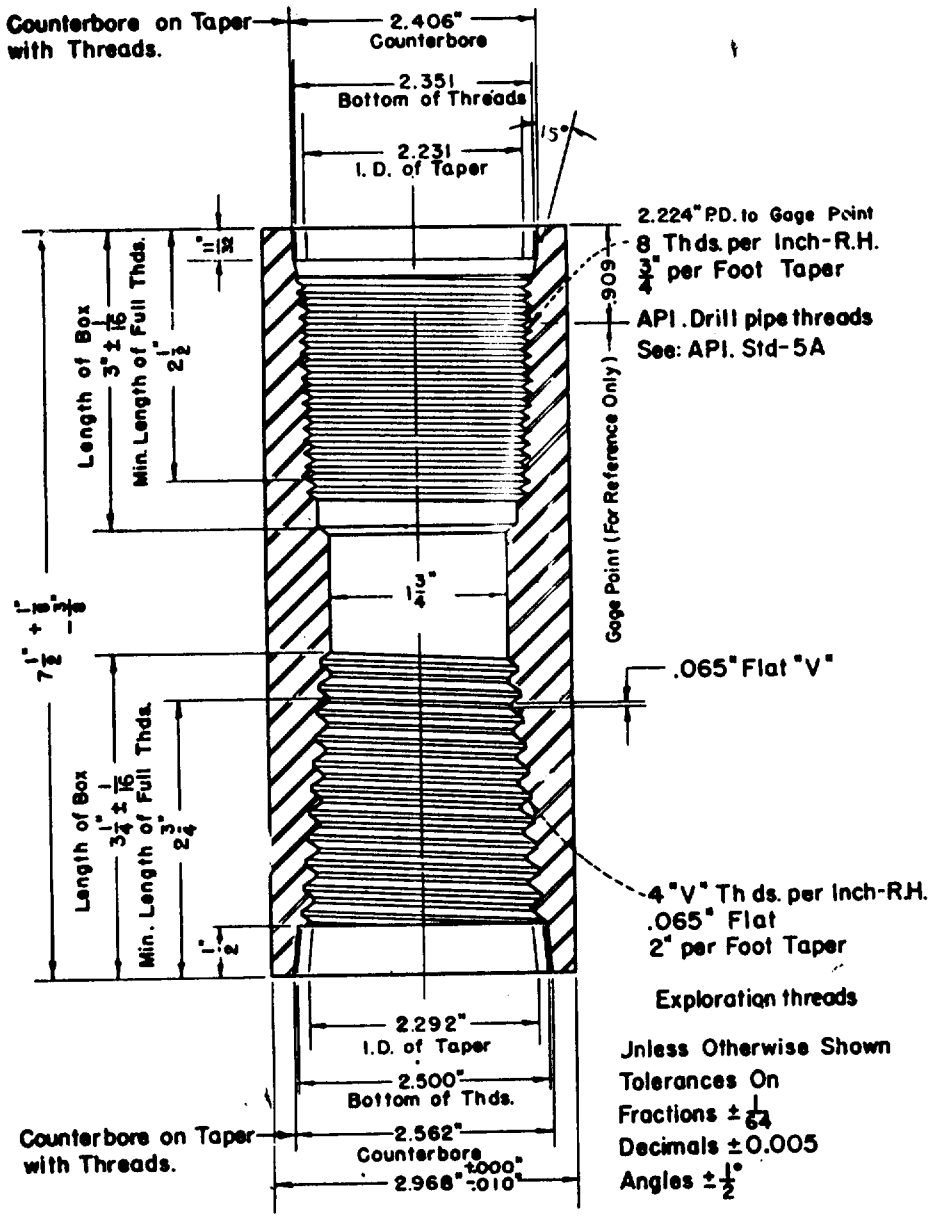
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Army - ME
Air Force - 99

Review activity:
Air Force - 84

Preparing activity:
Army - ME

Project 3820-0180

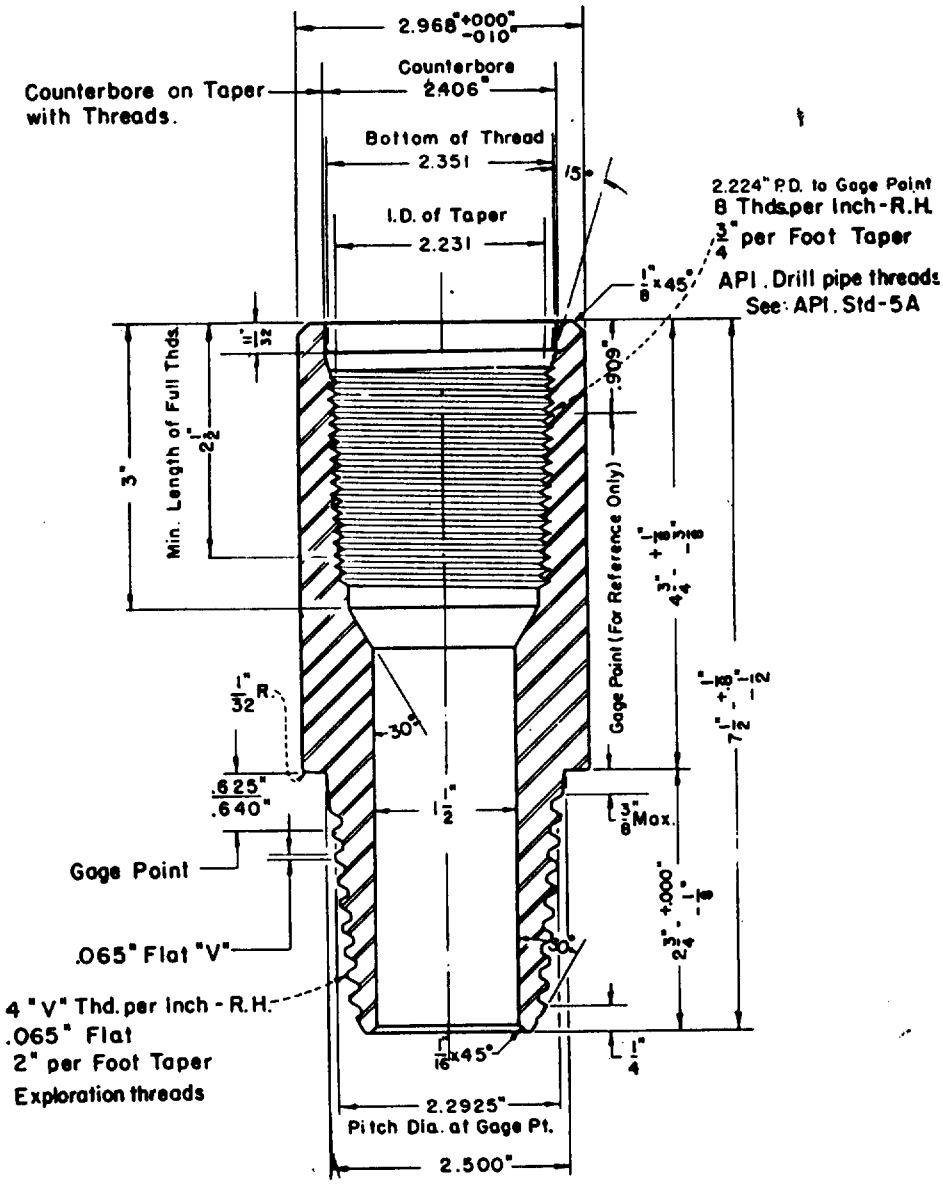
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MATERIAL: $3\frac{1}{16}$" O.D. x $1\frac{3}{4}$" I.D. A-4140 S.S. Tube H.R. x $7\frac{3}{4}$"
Tool Joint Stock Annealed & Straightened
Open Hearth - Random Lengths
O.D. Tolerance $\pm\frac{1}{32}$
Wall Tolerance ±10%
Heat Treatment: Y P 125,000 P.S.I. - Rockwell - C-31-34

FIGURE 1. Tool joint-female, 2-3/8 inch exploration drillpipe. X-3511

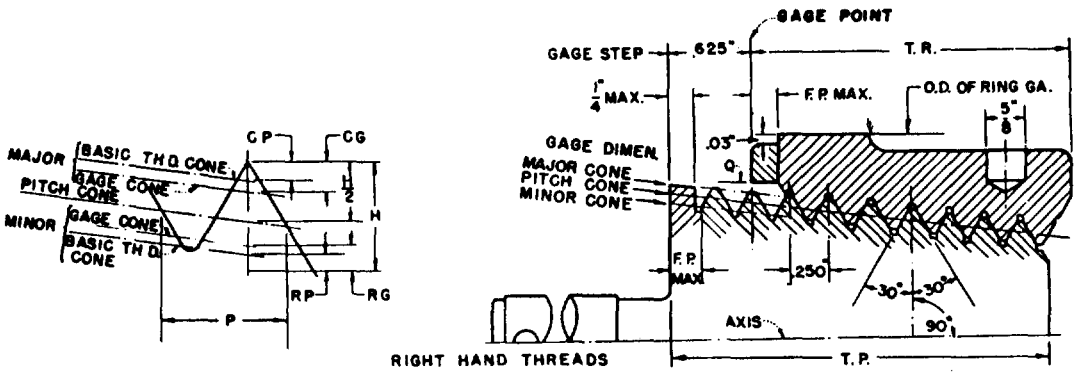
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MATERIAL: 3 1/16" O.D. x 1 1/2" I.D. A-4140 S.S. Tube H.R. x 7 3/4"
Tool Joint Stock Annealed & Straightened
Open Hearth - Random Lengths
O.D. Tolerance ± 1/32"
Wall Tolerance ±10%
Heat Treatment: Y.P. 125,000 P.S.I. - Rockwell - C 31 - 34

FIGURE 2. Tool joint-male, 2-3/8 inch exploration drillpipe. X-3512

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THREAD AND GAGE DATA

Size	No. Threads Per Inch	Pitch "P"	Taper Inches Per Foot	Height "H"	Thread Truncation: Product "CP" and "RP"	Thread Truncation: Gage "CG" and "RG"	Pitch Diameter at Gage Point	"TP"	"TR"	"OD"	"FP"	"Q"
2-3/8	4	.250	2	.21600	.05629	.06129	2.2925	2.75	2.125	3.75	3/8	2-1/2

GAGE TOLERANCES

Pitch Diam.	Lead Tolerance: Plug	Lead Tolerance: Ring	Taper (Difference in Diam. in TR Length) Nominal	Tolerance: Plug	Tolerance: Ring
±.004	±.0004	±.0006	.3541	+.0004 -.0000	+.0004 -.0012

All Dimensions in Inches at 68° Farenheit

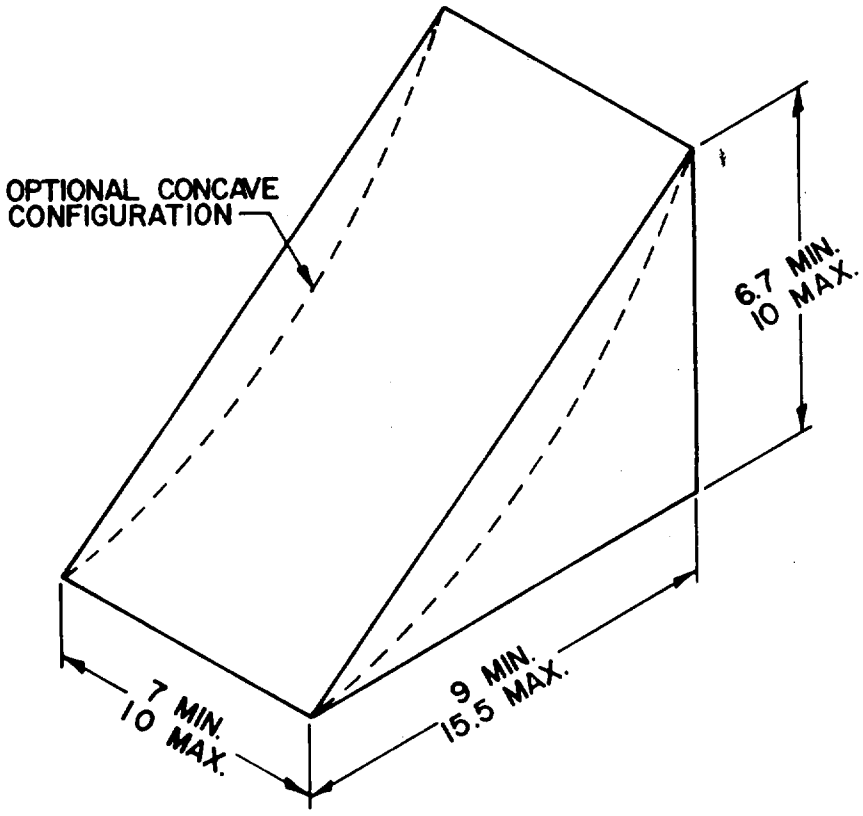
1. The gages shall be steel, hardened and ground. They may be manufactured with a fitting plate or ring as illustrated. The plug gage shall have a handle of a good commercial design and provision made to wrench both the plug and ring gages. (5/8" holes shown).
2. The first thread on both ends of the plug and ring gages shall be removed until the full form of the thread is obtained.
3. The plug gage in all cases shall be the standard.
4. The lead and the taper on plug and ring gages shall be measured along the pitch line over the full threaded portion, omitting one full thread from each end.
5. The ring gage shall be controlled by its engagement with the mating plug and the step value may vary between the limits 0.624" and 0.626".
6. The lead between any two threads within the lengths specified in paragraph 4 may vary from the nominal by not more than as shown in the above table.
7. The flanks of the thread may vary from the basic angle of 30 degrees to the normal by not more than the following values:

Plug Gages	7 min. of arc.
Ring Gages	15 min. of arc.

FIGURE 3. Thread and gages specifications for internal upset exploration drillpipe tool joints for 2-3/8 inch drillpipe size.

X-3513

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CHOCK MATERIAL NOTE

1. ALUMINIUM
2. SEMI-RIGID ELASTOMER
3. MAXIMUM CHOCK WEIGHT - 11.5 LB.

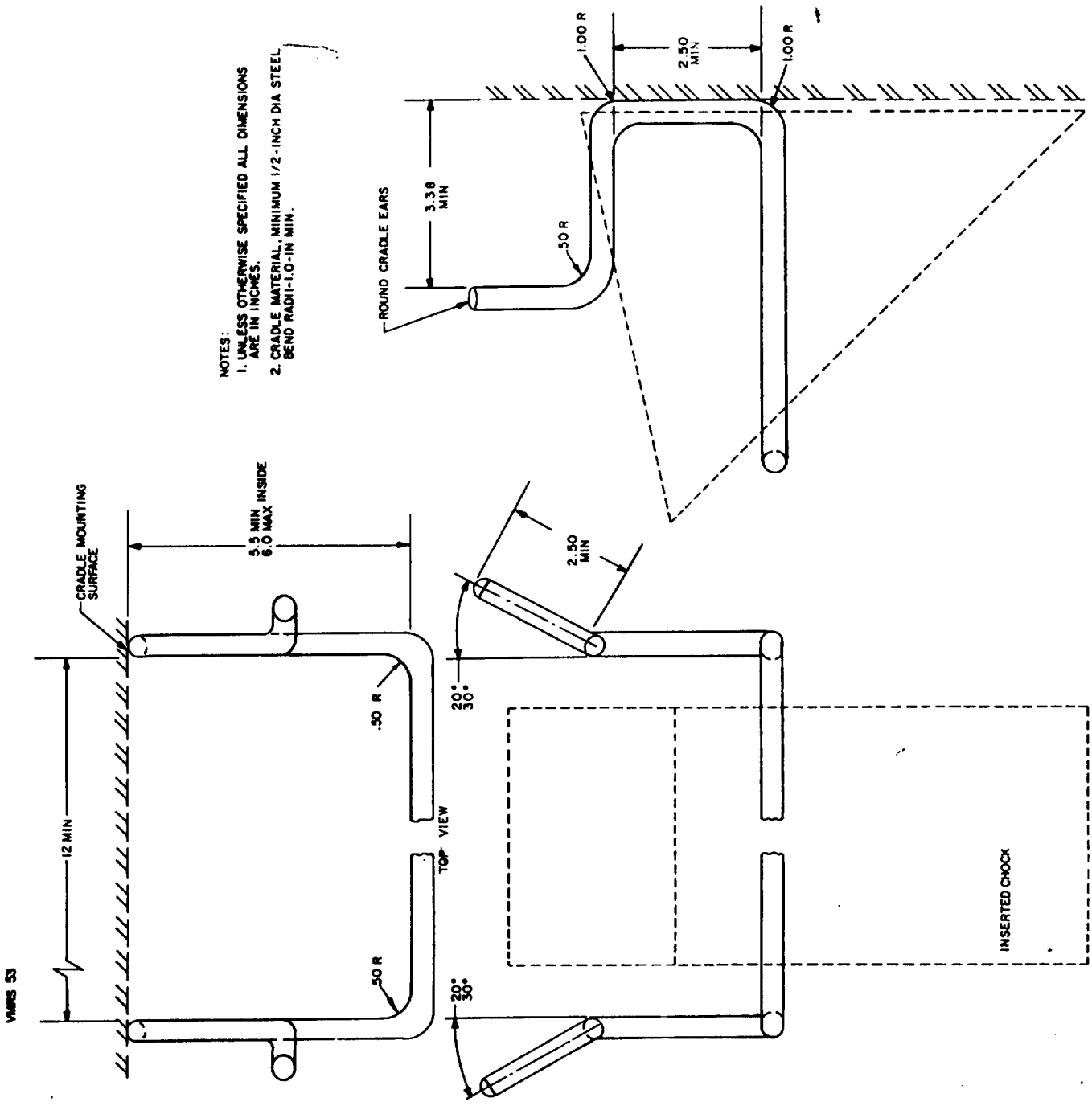
CAPTIVE FASTENER ASSEMBLY DETAILS

1. SNAP HOOK FOR CRADLE OR EYE.
2. FASTENER LINE TENSILE STRENGTH MINIMUM OF 1,400 LB.
3. PLATE FERROUS COMPONENTS TO RESIST CORROSION.
4. LENGTH OF CAPTIVE FASTENER ASSEMBLY SHALL REACH BETWEEN CRADLE OR HANGER AND EMPLACED CHOCK BLOCK AT EXTREME LOCATIONS AHEAD OF AND BEHIND ROAD WHEELS.

FIGURE 4. Chock block assembly.

X 603 B

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X-6048

FIGURE 5. Cradle.

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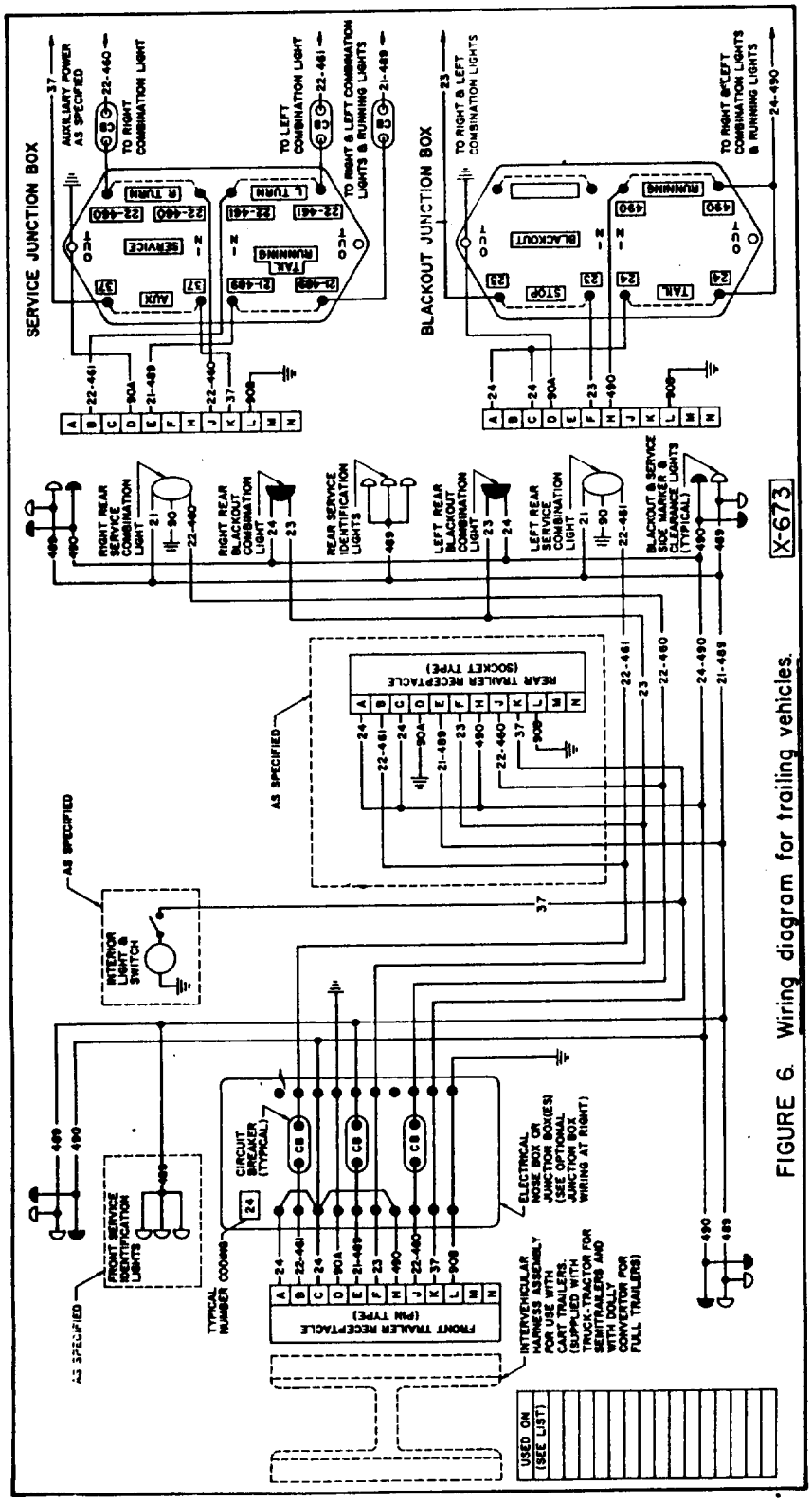


FIGURE 6. Wiring diagram for trailing vehicles.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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